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Bone fixation systems and methods

Abstract

A system and a bone fixation plate for preventing a fastener from rotating so as to prevent the fastener from backing out is provided. The system includes a plate having an opening. A retention member having an arm extending into the opening is configured to engage the head of the fastener so as to prevent the fastener from rotating after the fastener is seated within a vertebral body or implant. In one embodiment, the retention member engages a top surface of the head of the fastener. In another embodiment, the retention member engages a side wall of the head of the fastener.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 15/459,068, filed Mar. 15, 2017, which is a continuation-in-part of U.S. patent application Ser. No. 15/056,177, filed Feb. 29, 2016 (now U.S. Pat. No. 9,877,756), which is a divisional application of U.S. patent application Ser. No. 13/010,392, filed Jan. 20, 2011 (now U.S. Pat. No. 9,271,770), all of which are incorporated herein in their entireties by reference.

FIELD OF THE TECHNOLOGY

(1) The present disclosure generally relates to the field of spinal orthopedics, and more particularly to methods and systems for bone fixation plates for vertebrae.

BACKGROUND

(2) The spine is a flexible column formed of a plurality of bones called vertebrae. The vertebrae include a hollow cavity and essentially stack one upon the other, forming a strong column for support of the cranium and trunk of the body. The hollow core of the spine houses and protects the nerves of the spinal cord. The different vertebrae are connected to one another by means of articular processes and intervertebral, fibrocartilaginous bodies. The intervertebral bodies are also known as intervertebral disks and are made of a fibrous ring filled with pulpy material. The disks function as spinal shock absorbers and also cooperate with synovial joints to facilitate movement and maintain flexibility of the spine. When one or more disks degenerate through accident or disease, nerves passing near the affected area may be compressed and consequently irritated. The result may be chronic and/or debilitating back pain due to these spinal disorders.

(3) One procedure for treating spinal disorders involves using substantially rigid plates to hold vertebrae in desired spatial relationships and orientations relative to each other. During the procedure, the spine can be approached anteriorly or posteriorly. In either case, holes are drilled and tapped in at least two of the vertebrae, to receive screws or other fasteners used to secure the plate. The holes are accurately positioned with reference to openings formed through the plate. In some cases, the screws may be self-tapping. Typically, the plate is curved about its longitudinal axis to facilitate contiguous surface engagement of the plate with the vertebrae. With the plate maintained against the vertebrae, the fasteners are secured within the holes. As a result, the plate maintains the attached vertebrae in a desired spacing and orientation with respect to each other.

(4) One of the problems associated with bone fixation systems and methods is the tendency of screws or other fasteners to gradually work loose from the vertebrae after fixation of the bone plate. Slight shock or vibration of the vertebrae, due to walking, climbing stairs, or more vigorous activity by the patient following treatment increases this tendency, jeopardizing the integrity of fixation. Moreover, as the fasteners work loose, the outward protrusion of the heads over other components of the system can be a source of discomfort and present the risk of trauma to adjacent and surrounding soft tissue.

SUMMARY

(5) A bone fixation plate includes at least one opening formed in a plate that is configured to receive a fastener so as to secure the plate to a vertebral body and another opening configured to receive a fastener so as to secure the plate to an implant. A retention member extends into at least one of the openings and is configured to engage the fastener so as to prevent the fastener from rotating and becoming loose.

(6) In one embodiment, the retention member is configured to engage a top surface of the head of the fastener. In another embodiment, the retention member is configured to engage a side wall of the head of the fastener.

(7) In one embodiment, the plate includes a first opening spaced apart from a second opening, the first and second openings are configured to provide access for securing a fastener into a vertebral body. The plate further includes a third opening disposed between the first and second openings so as to provide access for securing a fastener to an implant.

(8) A system for bone fixation includes a plate including a first opening, a second opening, and an opening between the first and the second openings; a retention member extending into the opening of the plate that is elastically deformable from a rest position to first and second toggled positions; a first screw including a head configured to elastically deform the retention member to the first toggled position; and a second screw including a head configured to elastically deform the retention member to the second toggled position.

(9) In other features, the retention member further includes an arm extending from the plate into the opening and a first projection extending from the arm into the first opening. A second projection extends from the arm into the second opening.

(10) In yet other features, the first projection blocks a portion of the first opening and the second projection blocks a portion of the second opening when the retention member is in the rest position. The first projection is inside the opening of the plate when the retention member is in the first toggled position. The second projection is inside the opening of the plate when the retention member is in the second toggled position. The first screw forces the first projection away from the first opening in the first toggled position. The second screw forces the second projection away from the second opening in the second toggled position.

(11) A method for bone fixation includes the steps of inserting a first screw into a first opening of a plate to secure the plate to a vertebra, advancing the first screw to elastically deform a retention member formed in an opening of the plate, wherein the opening is in communication with the first opening, elastically deforming the retention member from a rest position to a first toggled position; and advancing the first screw until the retention member elastically returns to the rest position.

(12) In other features, elastically deforming the retention member from the rest position to the first toggled position includes flexing an arm of the retention member. Advancing the first screw to elastically deform the retention member includes contacting a projection of the retention member with a head of the first screw.

(13) In yet other features, the method further includes the steps of inserting a second screw into a second opening of the plate, advancing the second screw to elastically deform the retention member formed in the opening of the plate, wherein the opening is in communication with the second opening, elastically deforming the retention member from the rest position to a second toggled position; and advancing the second screw until the retention member elastically returns to the rest position. The first opening and the second opening are partially blocked when the retention member is in the rest position and one of the first and second openings is partially blocked when the retention member is in one of the first toggled position and the second toggled position.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is an elevation view of a bone fixation system attached to adjacent vertebrae according to the principles of the present disclosure.

(2) FIG. 2 is an exploded perspective view of the bone fixation system according to the principles of the present disclosure.

(3) FIG. 3 is a top elevation view of the bone fixation plate according the principles of the present disclosure.

(4) FIG. 4 is a bottom elevation view of the bone fixation plate according to the principles of the present disclosure.

(5) FIG. 5 is a partial perspective view of a retention mechanism in the bone fixation plate of FIG. 2 according to the principles of the present disclosure.

(6) FIGS. 6A-6D are partial cross-sectional views of a portion of the bone fixation plate along line VI of FIG. 3 and a bone screw according to the principles of the present disclosure.

(7) FIGS. 7A-7E are partial top elevation views of one end of the bone fixation system as the end is attached to one of the vertebrae.

(8) FIG. 8 is a perspective view of another bone fixation plate according to the principles of the present disclosure.

(9) FIG. 9 is a top view of another bone fixation plate according to the principles of the present disclosure.

(10) FIG. 10 is a perspective view of another embodiment of a fixation plate having a retention member configured to engage a top surface of a fastener.

(11) FIG. 11 is a top down view of the fixation plate shown in FIG. 10 showing the fastener engaged with the retention members.

(12) FIG. 12 is top down view of the fixation plate shown in FIG. 10 in isolation.

(13) FIG. 13A is a cross-sectional view of FIG. 11 showing the head of the fastener pushing down on the retention member.

(14) FIG. 13B is a view of FIG. 13A showing the fastener advancing.

(15) FIG. 13C is a view of FIG. 13B showing the retention member engaging the top surface of the fastener.

(16) FIG. 14 is a perspective view of another embodiment of the fixation plate with a retention member configured to prevent the fastener from rotating.

(17) FIG. 15 is a top down view of FIG. 14 showing the retention member biased radially by the head of the fastener.

(18) FIG. 16 is a top down view of FIG. 15 showing the retention member engaged with the

fastener.

DETAILED DESCRIPTION

(19) Some fasteners and bone fixation plates may include retention mechanisms that may prevent the screws from working loose from the plate and vertebra. However, many retention mechanisms are cumbersome and difficult to operate during surgical procedures. For example, some plates may have retention mechanisms that include additional hardware and features such as sliding tabs or projections, springs, and the like. The additional hardware adds weight and complexity to the plates and presents potential failure modes such as jammed tabs, broken springs, or other failures from the separate moving parts. The additional hardware may also require activation by the surgeon. That is, the surgeon may be required to find the activation point and twist or press a feature on the screw or plate to activate the retention mechanism. These additional steps require additional time and effort by the surgeon who is already under difficult working conditions such as low light and a confined surgical area.

(20) Accordingly, a bone fixation system and method are provided to decrease weight and complexity while reducing the duration of the surgical procedure. The bone fixation plate of the present disclosure includes an integrated retention mechanism that is formed in the bone fixation plate itself. The retention mechanism includes geometry introduced into the bone fixation plate during the manufacturing process. For example, the geometry may be introduced by forming the retention mechanism out of a mold or cutting with a laser or plasma cutting technique. Thus, the retention mechanism requires no additional hardware on the bone fixation plate. The bone fixation plate of the present disclosure further includes a self-activating retention mechanism that automatically locks when a fastener is inserted to attach the plate to a vertebra.

(21) Embodiments of the invention will now be described with reference to the Figures, wherein like numerals reflect like elements throughout. The terminology used in the description presented herein is not intended to be interpreted in any limited or restrictive way, simply because it is being utilized in conjunction with detailed description of certain specific embodiments of the invention. Furthermore, embodiments of the invention may include several novel features, no single one of which is solely responsible for its desirable attributes or which is essential to practicing the invention described herein.

(22) Referring now to FIG. 1, a bone fixation plate **100**, hereinafter referred to as the “plate,” is configured to be attached to a first vertebra (not shown) and a second vertebra (not shown) using a fastener, illustratively shown as bone screws **200**. The plate **100** may be substantially rectangular and include rounded sides and/or contoured surfaces to facilitate movement of tissue relative to the plate **100** after implantation in a patient. The plate **100** includes openings **104** formed in corners of the plate **100**. The screws **200** pass through the openings **104** to attach to the first and second vertebrae. Retention mechanisms **106**, formed between the openings **104** in the plate **100**, keep the screws **200** from backing out of the respective first and second vertebrae after fixation. A window **108** may also be formed in the plate **100** to provide access to the space between the first vertebra and the second vertebra. Although the system and method shown herein may be used to fix together two vertebrae, it is understood that the system and method may be applicable to multiple vertebrae in excess of two. For example, the plate **100** may extend beyond the first and/or second vertebra to attach to other adjacent vertebrae and may include additional retention mechanisms disposed between additional pairs of openings and screws.

(23) Referring now to FIG. 2, a first screw **200a** may be inserted through a first opening **104a** and a second screw **200b** may be inserted through a second opening **104b**. A first retention mechanism **106a** is disposed between the first and second openings **104a** and **104b** and is configured to prevent both the first screw **200a** and the second screw **200b** from backing out of the plate **100** after fixation to the first vertebra. In one embodiment, the first retention mechanism **106a** may be molded or machined from the material of the plate **100**. Thus, the first retention mechanism **106a** may be integral and continuous with the plate **100**. In other embodiments, the first retention

mechanism **106a** may be attached to the plate **100** between the openings **104a** and **104b**. The first retention mechanism **106a** may be attached by welding, snap fit, friction welding, or other forms of attachment.

(24) For simplicity, one retention mechanism and one pair of adjacent openings are discussed in detail herein. However, it is understood by one in the art that the plate **100** of the present disclosure may include one or more retention mechanisms disposed between any pair of openings. For example, third and fourth openings **104c** and **104d** may be mirror images of the first and second openings **104a** and **104b** and configured to receive third and fourth screws, respectively. Similarly, a second retention mechanism **106b** may be disposed between the third and fourth openings **104c** and **104d**. It may be readily understood by one in the art that the second retention mechanism **106b** functions the same as the first retention mechanism **106a** with respect to the third and fourth openings **104c** and **104d**. Thus, each of the retention mechanisms **106a** and **106b** may be referred to hereinafter as simply retention mechanism **106**. Likewise, each pair of openings may be referred to hereinafter as simply first and second openings **104a** and **104b**.

(25) Referring now to FIG. 3, a top view of the plate **100** illustrates additional features of the retention mechanism **106** and openings **104a** and **104b**. The first opening **104a** and the second opening **104b** are formed in one end of the plate **100**. The first and second openings **104a** and **104b** may be substantially circular and configured to receive the bone screws **200a** and **200b** respectively as shown in FIG. 2. The retention mechanism **106** is disposed in an opening **110** formed between the first and second openings **104a** and **104b**. The opening **110** may be irregular-shaped to conform to the geometry of the retention mechanism **106**. In one embodiment, the opening **110** may be in communication with both the first opening **104a** and the second opening **104b**. That is, a continuous wall is shared between the first opening **104a**, the second opening **104b**, and the opening **110**.

(26) The retention mechanism **106** extends from the plate **100** into the opening **110** and is configured to lock the bone screws **200a** and **200b** to the plate **100**. For example, the retention mechanism **106** may include a flexible arm **112** extending from the plate **100** into the opening **110**. The flexible arm **112** may be attached to the plate at a proximal end **114** and include a distal end **116** cantilevered in the opening **110**. A first projection **118a** extends from the distal end **116** through the opening **110** and towards the first opening **104a**. A second projection **118b** extends from the distal end **116** through the opening **110** towards the second opening **104b**. Thus, the retention mechanism **106** may resemble an inverted “T” shaped geometry or a nautical anchor. For example, the first and second projections **118a** and **118b** (collectively projections **118**) may include curvature having a radius approximately equivalent to the length of the arm **112** from the proximal end **114** to the distal end **116**.

(27) Referring now to FIG. 4, a bottom view of the plate **100** illustrates additional features that enable retention of two bone screws with one retention mechanism **106**. Each of the projections **118a** and **118b** includes a notch **120a** and **120b** respectively (collectively notches **120**), formed at the ends of the projections **118**. The notches **120** permit the projections **118** to slide over heads of the screws **200** after fixation to the vertebra as described below with reference to FIGS. 6A-6D and 7A-7E. The notches **120** may be configured to adjust the strength of the retention mechanism **106** by varying the thickness of the projections **118** and/or the arms **112**. For example, the notches **120** may decrease the thickness of the retention mechanism **106** and thus increase flexibility. Alternatively, the retention mechanism **106** may be of a uniform thickness that permits the projections **118** to slide over heads of the screws **200**.

(28) The retention mechanism **106** is capable of elastically moving, bending, or toggling between at least three distinct positions. In a rest position, as shown in FIGS. 1-5, the arm **112** “rests” substantially equidistant from the first and second openings **104a** and **104b**. Both the first projection **118a** and the second projection **118b** extend partially into the first and second openings **104a** and **104b** respectively to retain first and second screws **200a** and **200b** within the openings

104a and **104b**. From the rest position, the retention mechanism **106** may be toggled to a first toggled position and a second toggled position as discussed below.

(29) Referring now to FIG. 5, a portion of the retention mechanism **106b** of FIG. 2, and more particularly the first projection **118a** is shown in greater detail. The first projection **118a** extends through a mouth **120** in the opening **110** and into the first opening **104a**. The mouth **120** provides communication between the opening **110** and the opening **104a**. The first projection **118a** may include a top surface **122** that substantially conforms to a top surface **123** of the plate **100**. The first projection **118a** may include a taper **124** that decreases in thickness from the top surface **122** to a tip **126** on the distal end of the first projection **118a**. The taper **124** may facilitate engagement with the screw **200** as illustrated in FIGS. 6A-6D.

(30) Referring now to FIGS. 6A-6D, engagement between the screw **200** and the retention mechanism **106** are shown in greater detail. The screw **200** includes a head **202** and a threaded shaft **204**. The head **202** may be driven by a screwdriver such that the shaft **204** penetrates the vertebra (not shown) to secure the plate **100**. In FIG. 6A, the screw **200** is inserted into the first opening **104a**. As the screw **200** advances through the first opening **104a**, a bottom surface **206** of the head **202** moves towards the retention mechanism **106**. As the head **202** is driven, the shaft **204** engages more of the vertebra, and the bottom surface **206** begins to contact the taper **124** of the first projection **118a** as shown in FIG. 6B. The bottom surface **206** exerts a downward force on the retention mechanism **106**. The downward force pushes against the taper **124** which transfers a portion of the downward force to the retention mechanism **106**. The flexible arm **112** (not shown) bends due to the force, and the projection **118a** begins to move away from the first opening **104a**, as illustrated in detail in FIG. 6B.

(31) Continuing now with FIG. 6C, as the screw **200** continues to be driven into the vertebra, the bottom surface **206** passes by the taper **124** and a side wall **208** of the head **202** contacts the tip **126** of the projection **118a**. At this point, the retention mechanism **106** is in the first toggled position. In a toggled position, the arm **112** of the retention mechanism **106** may be fully bent towards the opposing opening as will be discussed in detail below. The screw **200** continues to be driven into the vertebra until fully seated in FIG. 6D. Once the screw **200** is fully seated, the retention mechanism **106** returns to the rest position. Now, the tip **126** of the projection **118a** extends over a top surface **210** of the head **202** of the screw **200** and the bottom surface **206** may contact a lip **127** in the opening **104a**. Thus, the retention mechanism **106** prevents the screw **200** from backing out from the plate **100**. Although the top surface **210** of the screw **200** contacts the projection **118a** in FIG. 6D, a gap may remain after the screw **200** attaches to a vertebra. As can be seen in FIG. 6D, a gap **128** may remain between the projection **118a** and the side wall **208** of the screw **200** once the retention mechanism **106** has returned to the rest position. The gap **128** permits the retention mechanism **106** to bend towards the first opening **104a** when the second screw **200b** is inserted into the second opening **104b** as discussed below with reference to FIGS. 7A-7E.

(32) Referring now to FIGS. 7A-7E, the bone fixation system and method are shown in greater detail by a series of partial top views. In FIG. 7A, a first half of the plate **100** is shown positioned over the first vertebra. The retention mechanism **106** is in the rest position with no screws in place. In FIG. 7B, the first screw **200a** is inserted into the first opening **104a** and driven into the vertebra. As described with respect to FIGS. 6B-C, the bottom surface **206** of the screw **200a** contacts the taper **124** causing the retention mechanism **106** to bend about the proximal end **114** of the arm **112**. The arm **112** elastically bends to the first toggled position before the first screw **200a** is fully seated in the first opening **104a**. In FIG. 7C, the first screw **200a** is fully seated in the first opening **104a**. The retention mechanism **106** returns to the rest position to prevent the first screw **200a** from backing away from the plate **100**. The tip **126** of the projection **118a** extends over the top surface **210** of the screw head **202**, as shown in FIG. 6D. Thus, the screw head **202** is locked between the projection **118a** and the lip **127**.

(33) Referring now to FIG. 7D, the second screw **200b** is inserted into the second opening **104b**

and driven into the vertebra. As described with respect to FIGS. 6B-C, the bottom surface **206** of the screw **200b** contacts the taper **124** on the distal end of the projection **118b** causing the retention mechanism **106** to bend about the proximal end **114** of the arm **112**. The arm **112** elastically bends to the second toggled position before the second screw **200b** is fully seated in the second opening **104b**. In FIG. 7E, the second screw **200a** is fully seated in the second opening **104b**. The retention mechanism **106** returns to the rest position to prevent the second screw **200b** from backing away from the plate **100**. The tip **126** of the projection **118b** extends over the top surface **210** of the screw head **202** of the second screw **202b**. Thus, the screw head **202** is locked between the projection **118b** and a corresponding lip **127** of the second opening **104b**.

(34) FIGS. **8** and **9** illustrate another embodiment of a bone fixation plate **800** that includes separate retention mechanisms **806a-d** (collectively retention mechanisms **806**) for each opening **804a-d** (collectively openings **804**). For example, a first retention mechanism **806a** is disposed between first and second openings **804a** and **804b** and is configured to prevent a first screw (not shown) in the first opening **804a** from backing out of the plate **800** after fixation to a vertebra. The first retention mechanism **806a** may be molded or machined from the material of the plate **800**. Thus, the first retention mechanism **806a** may be integral and continuous with the plate **800**. In other embodiments, the first retention mechanism **806a** may be attached to the plate **100** between the openings **804a** and **804b**. The first retention mechanism **806a** may be attached by welding, snap fit, friction welding, or other forms of attachment. A second retention mechanism **806b** is disposed between the second opening **804b** and the first retention mechanism **806a** and is configured to prevent a second screw (not shown) in the second opening **804b** from backing out of the plate **800** after fixation to the vertebra.

(35) For simplicity, two retention mechanisms and one pair of adjacent openings are discussed in detail herein. However, it is understood by one in the art that the plate **800** of the present disclosure may include two retention mechanisms disposed between any pair of openings. For example, third and fourth openings **804c** and **804d** may be mirror images of the first and second openings **804a** and **804b** and configured to receive third and fourth screws, respectively. Similarly, a third retention mechanism **806c**, may be disposed between the third and fourth openings **804c** and **804d** and a fourth retention mechanism **806d** may be disposed between the fourth opening **804d** and the third retention mechanism **806c**.

(36) Referring now to FIG. **9**, a top view of the plate **800** illustrates additional features of the retention mechanism **806** and openings **804a** and **804b**. The first opening **804a** and the second opening **804b** are formed in one end of the plate **100**. The first and second openings **804a** and **804b** may be substantially circular and configured to receive the bone screws **200a** and **200b** respectively (not shown), similar to openings **104a** and **104b** as shown in FIG. **2**. Each retention mechanism **806** is disposed in a separate opening **810** formed around each retention mechanism **806**. The openings **810** may be irregular-shaped to conform to the geometry of the retention mechanism **806**. In one embodiment, each opening **810** may be in communication with a neighboring opening **804**. That is, a continuous wall is shared between one opening, such as opening **804a** and one opening **810**.

(37) Each retention mechanism **806** extends from the plate **800** into the opening **810** and is configured to lock one bone screw to the plate **800**. The retention mechanism **806** includes a flexible arm **812** extending from the plate **800** into the opening **810**. The flexible arm **812** may be attached to the plate at a proximal end **814** and include a distal end **816** cantilevered in the opening **810**. A projection **818** extends from the distal end **816** through the opening **808** and towards the opening **804**. Thus in the present embodiment, the retention mechanism **806** may resemble a “J” shaped geometry or a cane. For example, the projection **818** may include curvature having a radius equivalent to the length of the arm **812** from the proximal end **814** to the distal end **816**.

(38) The retention mechanisms **806** function similar to the retention mechanism **106** as depicted in FIGS. **6A-6D**. That is, each retention mechanism **806** includes a taper on the projection **818** that comes in contact with a bottom surface of a screw when the screw is driven into a vertebra. As the

screw is driven further into the vertebra, the retention mechanism **806** contacts an outer wall of the head of the screw and moves from a rest position to a toggled position. However, in the present embodiment, each retention mechanism **806** locks one screw rather than two screws as retention mechanism **106**. Thus, the retention mechanism **806** in the present example may be toggled between the rest position as depicted in FIGS. **8** and **9** and one toggled position similar to the first toggled position of the retention mechanism **106**.

(39) With reference now to FIGS. **10-16** an aspect of the fixation plate is provided wherein the retention member is configured to engage the fastener so as to prevent the fastener from rotating. With reference first to FIG. **10** the fixation plate **300** is configured to be attached to a first vertebra (not shown) and a second vertebra (not shown) using a fastener, for example, and without limitation, bone screws **200** depicted in FIGS. **1-2**. The fixation plate **300** may be substantially rectangular and include rounded sides and/or contoured surfaces to facilitate movement of tissue relative to the plate **300** after implantation in a patient.

(40) The plate **300** is illustratively shown having a pair of vertebral openings **304** formed opposite ends of the plate **300**. The vertebral openings **304** are configured to allow the screws **200** to find purchase in a vertebral body. A central opening **306** is disposed between the vertebral openings **304**. The central opening **306** is configured to allow a screw **200** to find purchase in an implant **1000**.

(41) The plate **300** includes at least one retention member **308**. For illustrative purposes, the plate **300** includes a pair of retention members **308**. The retention members **308** are formed between the vertebral openings **304** and the central opening **306**. The retention members **308** are configured to engage the screws **200** so as to prevent the screws **200** from rotating and backing out of the respective vertebral openings **304** and the central opening **306**.

(42) FIG. **10**, shows a screw **200** being introduced into the central opening **306**, it should be appreciated that the screw **200** shown in FIG. **10** may be modified for use to find purchase in vertebral bodies. In particular, the screws illustratively shown in FIG. **11** all have the same head **202** but the shafts **204** of the screws may differ to accommodate the intended use. For instance, the thread pitch and shape of the screw shown in FIG. **10** is intended to couple with an implant. However, the thread pitch and shape of screws intended to find purchase in a vertebral body may differ.

(43) In one embodiment, the retention members **308** may be molded or machined from the material of the plate **300**. Thus, the retention members **308** may be integral and continuous with the plate **300**. In other embodiments, the retention members **308** may be formed separately from the plate **300** and attached to the plate **300** between the vertebral openings **304** and the central opening **306**. For instance, the retentions members **308** may be attached by welding, snap fit, friction welding, or other forms of attachment.

(44) For simplicity, the operation and structure of one retention member **308** and the central opening **306** and one vertebral opening **304** is discussed in detail herein. However, it should be appreciated by those skilled in the art that the discussion of one retention member **308** is sufficient to describe and enable the construction of the other retention members. It should be further appreciated that the embodiments described herein show two vertebral openings **304** and one central opening **306**, but the plate **300** may be modified to include additional openings without deviating from the scope of the appended claims. For instance, the plate **300** may have four openings as shown in FIGS. **1-4**.

(45) Referring again to FIG. **10** and also to FIG. **12**, a perspective view and a top down view, respectively, of the plate **100** illustrates additional features of the retention member **300**. The retention member **308** is a resilient material having a stem **308a** and an arm **308b**. The arm **308b** is generally orthogonal to the stem **308a**. The opposing sides of the stem **308a** is spaced apart from the body of the plate **300** so as to form two slots **300a** on each side of the stem **308a**. The slots **308a** allow the stem **308a** to flex. The arm **308b** extends into a respective opening. For illustrative

purposes, the retention member **308** is shown as having two arms **308b** extending outwardly and opposite each other into a respective vertebral opening **304** and central opening **306** so as to form the general shape of a “T”.

(46) The screw **200** includes a head **202** having a top surface **202a** configured to engage an arm **308b** of the retention member **308**. In one embodiment top surface **202a** includes a plurality of nubs **202b** spaced apart from each other so as to form detents **202c**. The arms **308b** may include a taper **308c**. The taper **308c** may facilitate the introduction of the screw **200** into the vertebral body or the implant, as the case may be, by facilitating the biasing of the retention member **308** about the stem **308a**. The screws include a shaft **204** with a thread **204a**. As described above, the thread **204a** is shown configured to couple with an implant, but may be modified to find purchase with a vertebral body.

(47) FIG. **11** shows each of the openings **304**, **306** with screws **200** disposed therein. In particular, each arm **308b** is seated in a respective detent **202c** and held therein by surrounding nubs **202**. Accordingly, the arms **308b** are engaged with a top surface **202a** of a respective screw **200** so as to prevent the screw from rotating and thus backing out of the implant or vertebral body as the case may be.

(48) With reference now to FIGS. **13A-13C**, an operation of the plate **300** is provided. Each end of the plate **300** may be mounted over adjacent vertebral bodies (not shown). As the operation of the retention members **308** is the same for the screws **200** in the vertebral openings **304** as the central opening **306** an illustration describing the central opening **306** is sufficient. FIG. **13A** shows the head **202** of the screw **200** biasing the retention member **308** as the screw **200** advances into the central opening **306**. As shown, both retention members **308** are biased away from the head **202**. FIG. **13B** shows the screw **200** advanced further. It should be appreciated that advancement of the screw **200** is done by turning the screw **200** so as to threadingly engage a reciprocating bore of an implant **1000**. FIG. **13C** shows the head **202** of the screw **200** seated against the implant **1000**. As the head **202** is turned, the arms **308b** of opposing retention members **308** are seated in a corresponding detent **202c** so as to engage the head **202** of the screw **200** to prevent the screw **200** from rotating.

(49) FIGS. **13A-13C** also show how the depth of the openings **304**, **306** are dimensioned so as to accommodate the head **202** of the screw **200**. Thus, the head **202** of the screw **200** may be advanced past the arm **308b** of the respective retention member **300**. Though FIGS. **13A-13C** show the central opening **306** registered to a threaded bore of an implant, the spatial dimensions illustrated may be applied to the vertebral openings **304** with respect to a vertebral body.

(50) With reference now to FIGS. **14-16** another aspect of the fixation plate **300** and the retention member **408** as a resilient member disposed within an opening of the plate **300** is shown. It is understood that a cut-away view of the plate **300** is shown in FIGS. **14-16**. FIGS. **14-16** provide isolated views of the opening **306** and the screw **200** to illustrate the operation of another aspect of the fixation plate **300** described above. As seen, a retention member **400** is illustratively shown as a curved arm having a flattened head. The proximal end of the curved arm is illustratively shown integrally formed to the body of the plate surrounding a respective opening. For illustrative purposes, the opening is shown as a central opening **406**, however, the retention member **400** may also be used in a vertebral opening **406**.

(51) With reference first to FIG. **14**, the retention member **400** includes an arm **400a** which is curved so as to generally follow a portion of the central opening **406**. A relief opening **404** is disposed adjacent the proximal end of the arm **400a** so as to provide tolerance for biasing the arm **400a**. In one embodiment, the relief opening **404** is a cylindrical bore extending between a top surface and bottom surface of the plate **300**. The arm **400a** includes an interior surface directly opposite the body of the plate **300** so as to be spaced apart from the body of the plate **300**. The interior surface of the arm **400a** is exposed to the central opening **406**.

(52) FIG. **15** also provides an illustrative embodiment of a screw **200** for engagement with the

retention member **400**. The screw **200** includes a head **202** having a side wall **202a**. A portion of the side wall **202a** includes a detent **202b**. The detent **202b** includes a flattened planar surface which is recessed with respect to the side wall **202a** so as to define an upper lip **202c** spaced apart from a lower lip **202d**. The detent **202b** is configured to engage a flattened head **400b** of the arm **400a** of the retention member **400**.

(53) FIG. **14** shows the retention member **400** in a natural state. FIG. **15** shows the retention member **400** biased against the inner wall of the plate **300** defining the central opening **306**. In particular, as the screw **200** is advanced into the central opening **306** by rotation and engagement with a threaded bore of an implant (such as is generally shown in FIGS. **13A-13C**), the head **202** of the screw **200** pushes the retention member **400** against the inner wall of the plate **300**. FIG. **15** shows the detent **202b** of the screw **200** offset from the flattened head **400b** of the retention member **400**. FIG. **16** shows the screw being advance further, wherein upon further rotation the flattened head **400b** of the retention member **400** is engaged with the detent **202b** of the screw **200**. The plate **300** may have a central opening **306** with a depth similar to that shown in FIGS. **13A-13C** so as to register the flattened head **400b** of the retention member **400** with the detent **202b** of the screw **200** when the screw **200** is seated against the implant.

(54) With reference again to FIGS. **10-16** a system **2000** for use in a spinal procedure is provided. The system **2000** configured to prevent a fastener from backing out. The system **2000** includes a fixation plate having a retention member configured to engage the fastener, illustratively shown as a screw, so as to prevent the fastener from rotating and backing out.

(55) FIGS. **10-13C** illustrate one embodiment of the system **2000** wherein the fixation plate **300** is configured to be attached to a first vertebra (not shown) and a second vertebra (not shown) using a fastener, illustratively shown as bone screws **200**. The fixation plate **300** may be substantially rectangular and include rounded sides and/or contoured surfaces to facilitate movement of tissue relative to the plate **300** after implantation in a patient.

(56) The plate **300** is illustratively shown having a pair of vertebral openings **304** formed opposite ends of the plate **300**. The vertebral openings **304** are configured to allow the screw **200** to find purchase in a vertebral body. A central opening **306** is disposed between the vertebral openings **304**. The central opening **306** is configured to allow a screw **200** to find purchase in an implant **1000**.

(57) The plate **300** includes at least one retention member **308**. For illustrative purposes, the plate **300** includes a pair of retention members **308**. The retention members **308** are formed between the vertebral openings **304** and the central opening **306**. The retention members **308** are configured to engage the screws **200** so as to prevent the screws **200** from rotating and backing out of the respective vertebral openings **304** and the central opening **306**.

(58) FIG. **10** shows a screw **200** being introduced into the central opening **306**. It should be appreciated that the screw **200** shown in FIG. **10** may be modified for use to find purchase in vertebral bodies. In particular, the screws illustratively shown in FIG. **11** all have the same head **202** but the shafts **204** of the screws may differ to accommodate the intended use. For instance, the thread pitch and shape of the screw shown in FIG. **10** is intended to couple with an implant. However, the thread pitch and shape of screws intended to find purchase in a vertebral body may differ.

(59) Referring again to FIG. **10** and also to FIG. **12**, a perspective view and a top down view, respectively, of the plate **100** illustrates additional features of the retention member **300**. The retention member **308** is a resilient material having a stem **308a** and an arm **308b**. The opposing sides of the stem **308a** is spaced apart from the body of the plate **300** so as to form two slots **300a** on each side of the stem **308a**. The slots **308a** allow the stem **308a** to flex. The arm **308b** extends into a respective opening. For illustrative purposes, the retention member **308** is shown as having two arms **308b** extend into a respective vertebral opening **304** and central opening **306** so as to form the general shape of a “T”.

(60) The screw **200** includes a head **202** having a top surface **202a** configured to engage an arm **308b** of the retention member **308**. In one embodiment top surface **202a** includes a plurality of nubs **202b** spaced apart from each other so as to form detents **202c**. The arms **308b** may include a taper **308c**. The taper **308c** may facilitate the introduction of the screw **200** into the vertebral body or the implant, as the case may be, by facilitating the biasing of the retention member **308** about the stem **308a**. The screws include a shaft **304** with a thread **304a**. As described above, the thread **304a** is shown configured to couple with an implant, but may be modified to find purchase with a vertebral body.

(61) The retention members **308** are configured to engage the top surface **202a** of the screw **200** so as to prevent the screw **200** from rotating. In one embodiment, the arm **308b** of the retention member **308** is configured to be seated in a detent **202c** so as to be nestled between adjacent nubs **202b** wherein the rotation of the screw **200** is prevented by the nubs **202b** restricting the movement of the arm **308b**.

(62) Another embodiment of the system **2000** is illustratively shown in FIGS. **14-16**, wherein the retention member **400** is illustratively shown as a curved arm having a flattened head. The proximal end of the curved arm is illustratively shown integrally formed to the body of the plate surrounding a respective opening. For illustrative purposes, the opening is shown as a central opening **406**, however, the retention member **400** may also be used in a vertebral opening **406**.

(63) With reference first to FIG. **14**, the retention member **400** includes an arm **400a** which is curved so as to generally follow a portion of the central opening **406**. A relief opening **404** is disposed adjacent the proximal end of the arm **400a** so as to provide tolerance for biasing the arm **400a**. In one embodiment, the relief opening **404** is a cylindrical bore extending between a top surface and bottom surface of the plate **300**. The arm **400a** includes an interior surface directly opposite the body of the plate **300** so as to be spaced apart from the body of the plate **300**. The interior surface of the arm **400a** is exposed to the central opening **406**.

(64) FIG. **15** also provides an illustrative embodiment of a screw **200** for engagement with the retention member **400**. The screw **200** includes a head **202** having a side wall **202a**. A portion of the side wall **202a** includes a detent **202b**. The detent **202b** includes a flattened planar surface which is recessed with respect to the side wall **202a** so as to define an upper lip **202c** and lower lip **202d**. The detent **202b** is configured to engage a flattened head **400b** of the arm **400a** of the retention member **400**.

(65) FIG. **14** shows the retention member **400** in a natural state. FIG. **15** shows the retention member **400** biased against the inner wall of the plate **300** defining the central opening **306**. In particular, as the screw **200** is advanced into the central opening **306** by rotation and engagement with a threaded bore of an implant (such as is generally shown in FIGS. **13A-13C**), the head **202** of the screw **200** pushes the retention member **400** against the inner wall of the plate **300**. FIG. **15** shows the detent **202b** of the screw **200** offset from the flattened head **400b** of the retention member **400**. FIG. **16** shows the screw being advance further, wherein upon further rotation the flattened head **400b** of the retention member **400** is engaged with the detent **202b** of the screw **200**. The plate **300** may have a central opening **306** with a depth similar to that shown in FIGS. **13A-13C** so as to register the flattened head **400b** of the retention member **400** with the detent **202b** of the screw **200** when the screw **200** is seated against the implant.

(66) Example embodiments of the methods and systems of the present invention have been described herein. As noted elsewhere, these example embodiments have been described for illustrative purposes only, and are not limiting. Other embodiments are possible and are covered by the invention. Such embodiments will be apparent to persons skilled in the relevant art(s) based on the teachings contained herein. Thus, the breadth and scope of the present invention should not be limited by any of the above-described exemplary embodiments, but should be defined only in accordance with the following claims and their equivalents.

Claims

1. A bone fixation system comprising: first and second fasteners, each fastener comprising a shank portion and a head portion; a plate having first and second sides, the plate comprising: a body portion having a length extending from a caudal end to a cephalad end, a first opening for receiving the first fastener, a second opening for receiving the second fastener, the first and second openings positioned next to each other, but not in communication with each other, along a line that is normal to the length of the body portion, a third opening extending from the first side to the second side, the third opening defined by a first pair of sidewalls substantially defining a line tangential to the first opening, a first retention member positioned between the first pair of sidewalls, the first retention member comprising a substantially straight stem portion extending from the body portion and an extension portion that extends from the stem portion in the third opening into the first opening, the stem portion being elastically deformable so that, at rest, the first retention member occupies a first position and, when deformed, occupies a second position, a fourth opening extending from the first side to the second side, the fourth opening defined by a second pair of sidewalls substantially defining a line tangential to the second opening, and a second retention member positioned between the second pair of sidewalls, the second retention member comprising a substantially straight stem portion extending from the body portion and an extension portion that extends from the stem portion in the fourth opening into the second opening, the stem portion being elastically deformable so that, at rest, the second retention member occupies a first position and, when deformed, occupies a second position, wherein the first and third openings are in communication, the second and fourth openings are in communication, and the third and fourth openings are not in communication; wherein insertion of the first fastener through the first opening causes the stem of the first retention member to at least partially deform when the head of the first fastener presses against the extension of the first retention member causing the first retention member to occupy the second position thereby allowing the head to pass and, when the head of the first fastener is advanced past the extension portion of the first retention member, the first retention member returns to the first position with at least a portion of the extension portion covering at least a portion of the head of the first fastener; wherein insertion of the second fastener through the second opening causes the stem of the second retention member to at least partially deform when the head of the second fastener presses against the head of the second retention member causing the second retention member to occupy the second position thereby allowing the head to pass and, when the head of the second fastener is advanced past the extension portion of the second retention member, the second retention member returns to the first position with at least a portion of the extension portion covering at least a portion of the head of the second fastener; and a fifth opening positioned next to, but not in communication with, the first opening along the length of the body portion.

2. The bone fixation system of claim 1, wherein the plate further comprises a sixth opening centrally located on the plate, the sixth opening not in communication with any of the first, second, third, or fourth openings.

3. The bone fixation system of claim 2, wherein the first, second, third, and fourth openings are positioned between the sixth opening and either the caudal or cephalad ends of the body portion.

4. The bone fixation system of claim 1, wherein the extension portion of each retention member defines a radius of curvature.

5. The bone fixation system of claim 4, wherein the radius of curvature is substantially normal to the stem portion of each retention member.

6. The bone fixation system of claim 1, wherein each extension portion comprises a proximal end connected to the stem portion, a distal end opposite the proximal end, and a ledge at the distal end, the ledge having a thickness that is less than a thickness of the rest of the extension portion.

7. The bone fixation system of claim 6, wherein the thickness of the rest of the extension portion is substantially the same as a thickness of the body portion of the plate.

8. A bone fixation system comprising: first and second fasteners, each fastener comprising a shank portion and a head portion; a plate having first and second sides, the plate comprising: a body portion having a length extending from a caudal end to a cephalad end, a first opening for receiving the first fastener, a second opening for receiving the second fastener, the first and second openings positioned along a line that is normal to the length of the body portion, a third opening extending from the first side to the second side, the third opening defined by a first pair of sidewalls substantially defining a line tangential to the first opening, a first retention member positioned between the first pair of sidewalls, the first retention member comprising a substantially straight stem portion extending from the body portion and an extension portion that extends from the stem portion in the third opening into the first opening, the stem portion being elastically deformable so that, at rest, the first retention member occupies a first position and, when deformed, occupies a second position, a fourth opening extending from the first side to the second side, the fourth opening defined by a second pair of sidewalls substantially defining a line tangential to the second opening, and a second retention member positioned between the second pair of sidewalls, the second retention member comprising a substantially straight stem portion extending from the body portion and an extension portion that extends from the stem portion in the fourth opening into the second opening, the stem portion being elastically deformable so that, at rest, the second retention member occupies a first position and, when deformed, occupies a second position, wherein the first and third openings are in communication, the second and fourth openings are in communication, and the third and fourth openings are not in communication; wherein insertion of the first fastener through the first opening causes the stem of the first retention member to at least partially deform when the head of the first fastener presses against the extension of the first retention member causing the first retention member to occupy the second position thereby allowing the head to pass and, when the head of the first fastener is advanced past the extension portion of the first retention member, the first retention member returns to the first position with at least a portion of the extension portion covering at least a portion of the head of the first fastener; wherein insertion of the second fastener through the second opening causes the stem of the second retention member to at least partially deform when the head of the second fastener presses against the head of the second retention member causing the second retention member to occupy the second position thereby allowing the head to pass and, when the head of the second fastener is advanced past the extension portion of the second retention member, the second retention member returns to the first position with at least a portion of the extension portion covering at least a portion of the head of the second fastener; wherein the plate further comprises a fifth opening centrally located on the plate, the fifth opening not in communication with any of the first, second, third, or fourth openings; and wherein the first, second, third, and fourth openings are positioned between the fifth opening and either the caudal or cephalad ends of the body portion.
